(a) Sz.-Nagy Unitary Dilation & Ancilla Post-Selection Architecture

Non − Hermitian Operator : $\mathcal{H}_{\text{AEGIS}} = \mathcal{W} + \mathbf{V} - i\mathbf{\Gamma}$

Contractive Scaling : $K = e^{-i\mathcal{H}_{\text{AEGIS}}\Delta t} / (\|\exp(-i\mathcal{H}\Delta t)\|_2 \cdot 1.05)$

Sz. − Nagy Unitary Dilation Operator ($U_{\text{dil}} \in \mathcal{U}(2N)$) :

$U_{\text{dil}} = [\ [K,\ D_{K^\dagger}],\ [D_K,\ -K^\dagger]\ ]$

where $D_K = \sqrt{I - K^\dagger K}$, $D_{K^\dagger} = \sqrt{I - KK^\dagger}$

Circuit Topology ($N = 4 \to 2$ System + 1 Ancilla) :

$|q_0\rangle_S$ − − − H − − − [U_{dil}] − − − Measure System $\to P(u)$

$|q_1\rangle_S$ − − − H − − − [(8×8)] − − − Measure System

$|a_0\rangle_A$ − − − $|0\rangle$ − − − [ ] − − − Post-Select $|0\rangle_A$ ($P_{\text{succ}} \approx$ 15-60%)





